

AZT-Machine Learning For Prediction

Group members:

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Model 1: Logistic Regression

Evaluation metrics: LR-Accuracy

Classification Report:

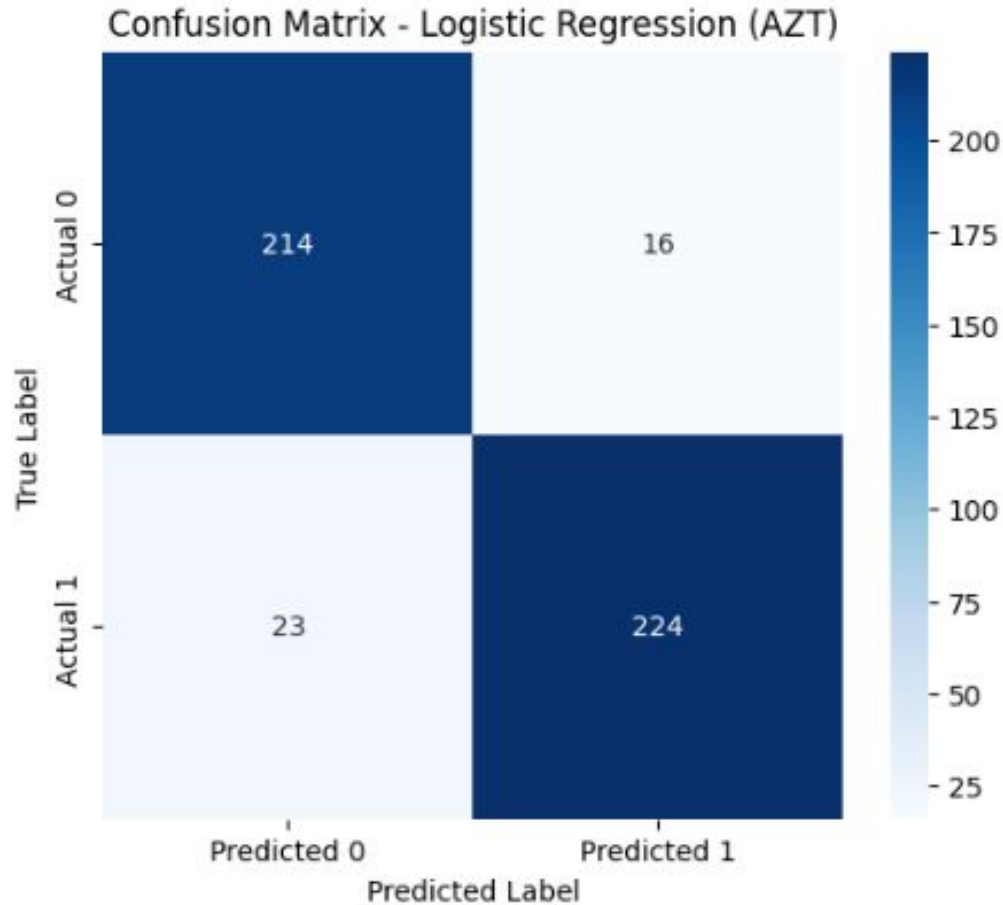
	precision	recall	f1-score	support
0	0.90	0.93	0.92	230
1	0.93	0.91	0.92	247
accuracy			0.92	477
macro avg	0.92	0.92	0.92	477
weighted avg	0.92	0.92	0.92	477

Evaluation metrics: LR-Accuracy

Classification Report:

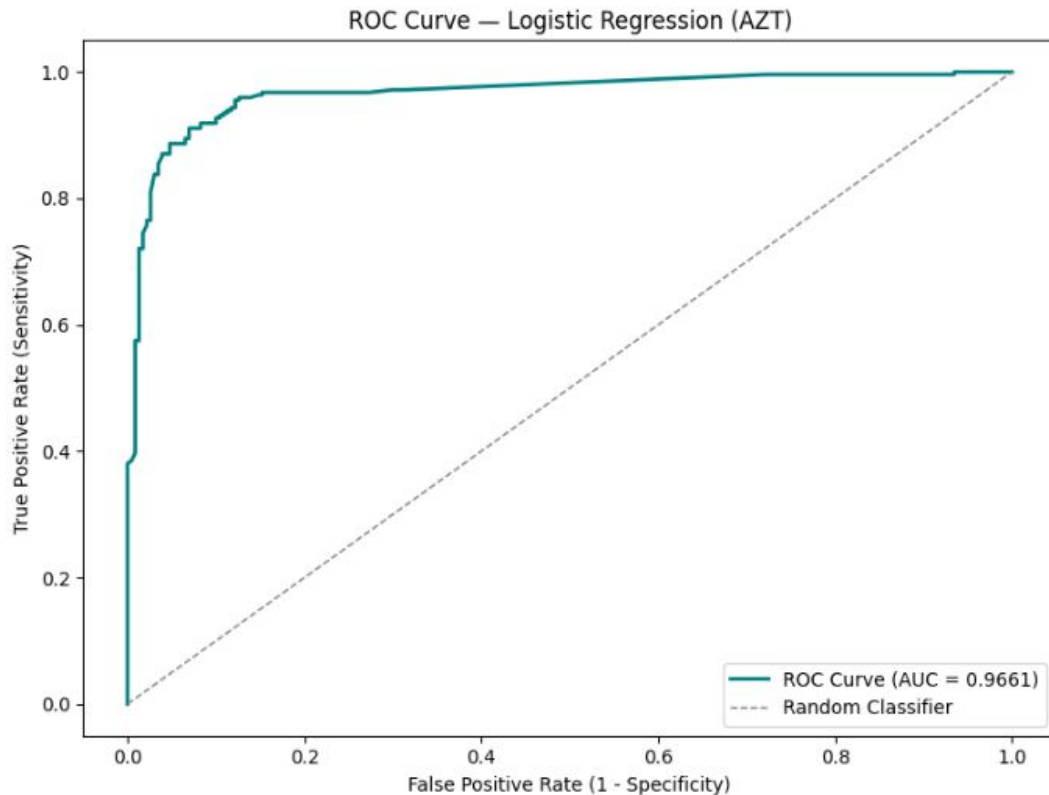
	precision	recall	f1-score	support
0	0.90	0.93	0.92	230
1	0.93	0.91	0.92	247
accuracy			0.92	477
macro avg	0.92	0.92	0.92	477
weighted avg	0.92	0.92	0.92	477

Confusion Matrix - Logistic Regression



Precision, Recall, F1 score, precision- Recall & **ROC AUC curves**

- Data set is balanced - (our dataset is imbalanced)
- False positives and False negatives are of equal importance #Critical for our model



Precision- Recall & ROC AUC curves

Model 2: Random Forest Classifier

Evaluation metrics: RF-Accuracy

--- Classification Report ---

	precision	recall	f1-score	support
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0	0.92	0.95	0.93	230
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1	0.95	0.92	0.94	247
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accuracy			0.94	477
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macro avg	0.93	0.94	0.93	477
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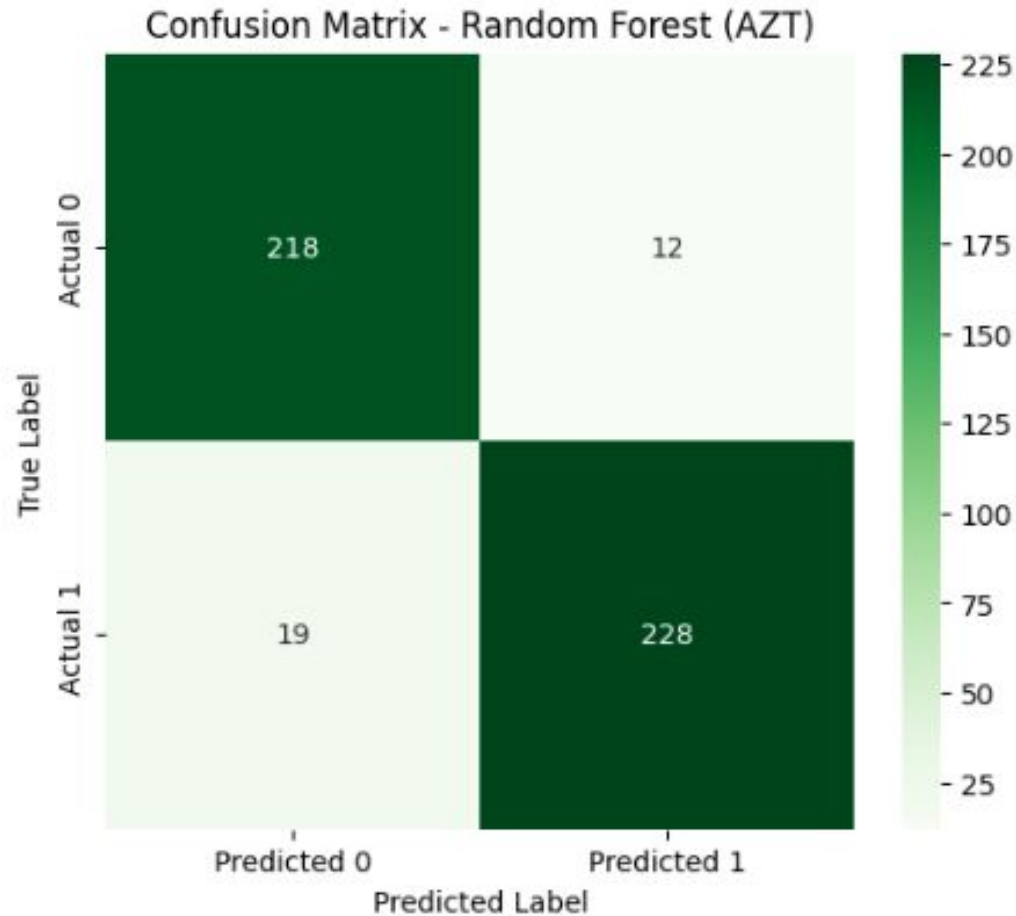
weighted avg	0.94	0.94	0.94	477
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Evaluation metrics: RF-Accuracy

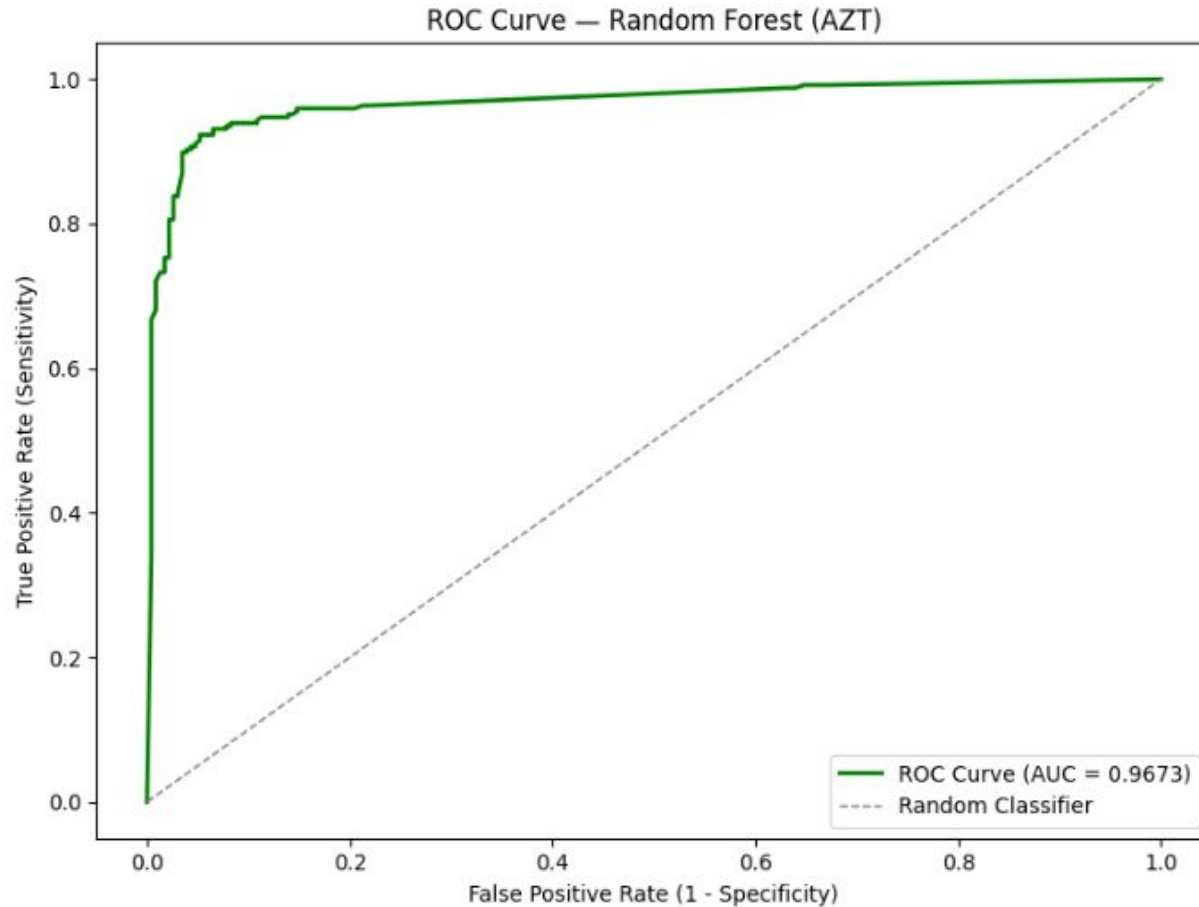
--- Classification Report ---

	precision	recall	f1-score	support
0	0.92	0.95	0.93	230
1	0.95	0.92	0.94	247
accuracy			0.94	477
macro avg	0.93	0.94	0.93	477
weighted avg	0.94	0.94	0.94	477

Confusion Matrix - Random Forest



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves

Model 3: Support Vector Machine

Evaluation metrics: SVM-Accuracy

--- SVM Test Classification Report ---

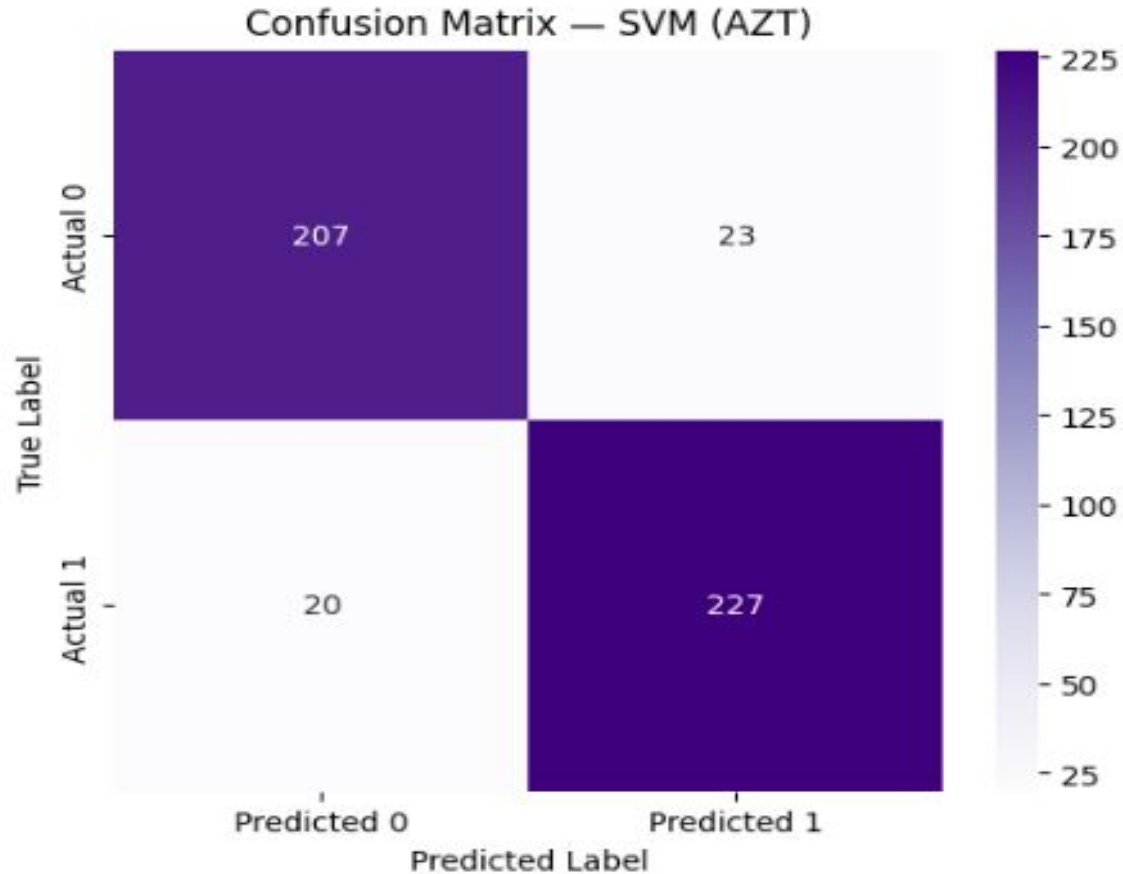
	precision	recall	f1-score	support
0	0.91	0.90	0.91	230
1	0.91	0.92	0.91	247
accuracy			0.91	477
macro avg	0.91	0.91	0.91	477
weighted avg	0.91	0.91	0.91	477

Evaluation metrics: SVM-Accuracy

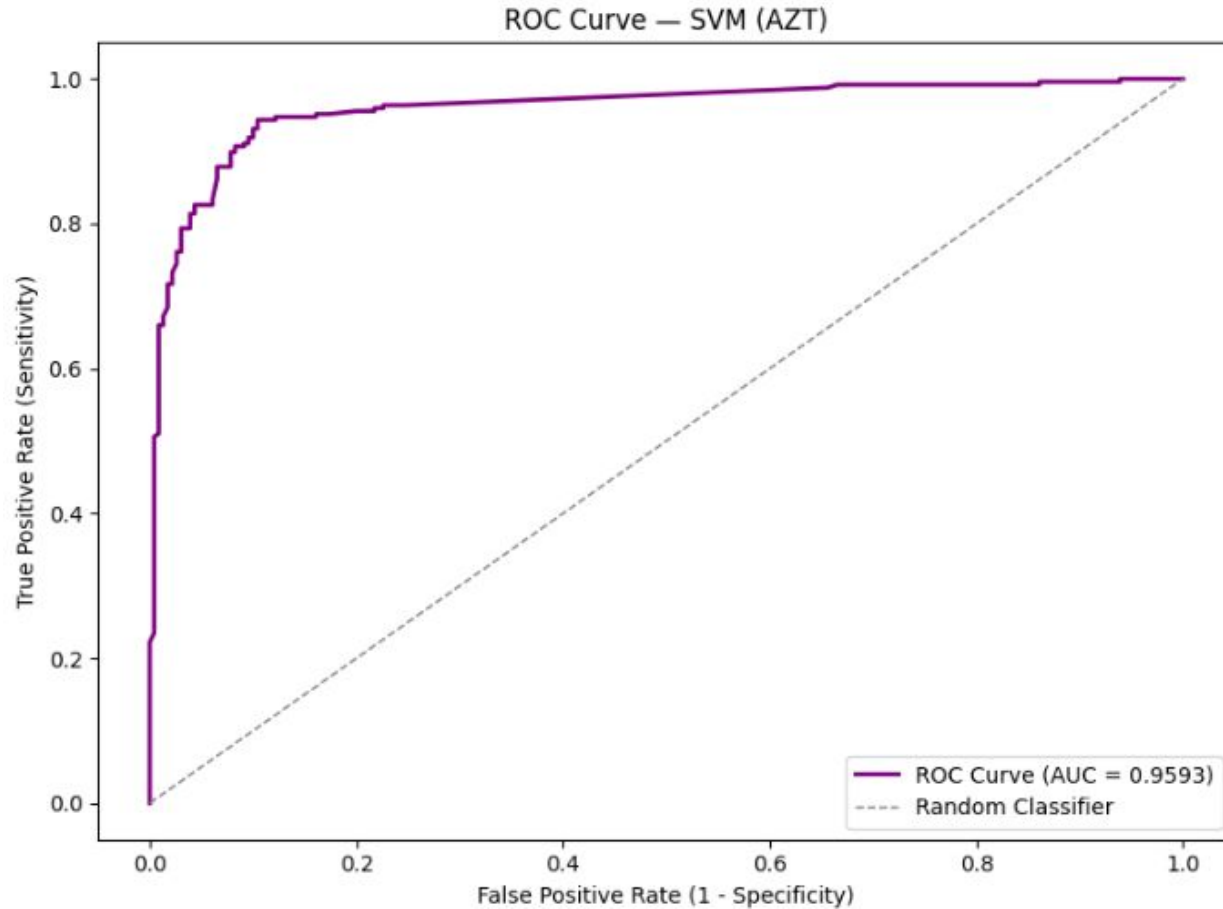
--- SVM Test Classification Report ---

	precision	recall	f1-score	support
0	0.91	0.90	0.91	230
1	0.91	0.92	0.91	247
accuracy			0.91	477
macro avg	0.91	0.91	0.91	477
weighted avg	0.91	0.91	0.91	477

Confusion Matrix - Support Vector Machine



Precision- Recall & ROC AUC curves



Precision- Recall & ROC AUC curves

Model 4: Extreme Gradient Boost

Evaluation metrics: XGB-Accuracy

--- XGBoost Test Classification Report ---

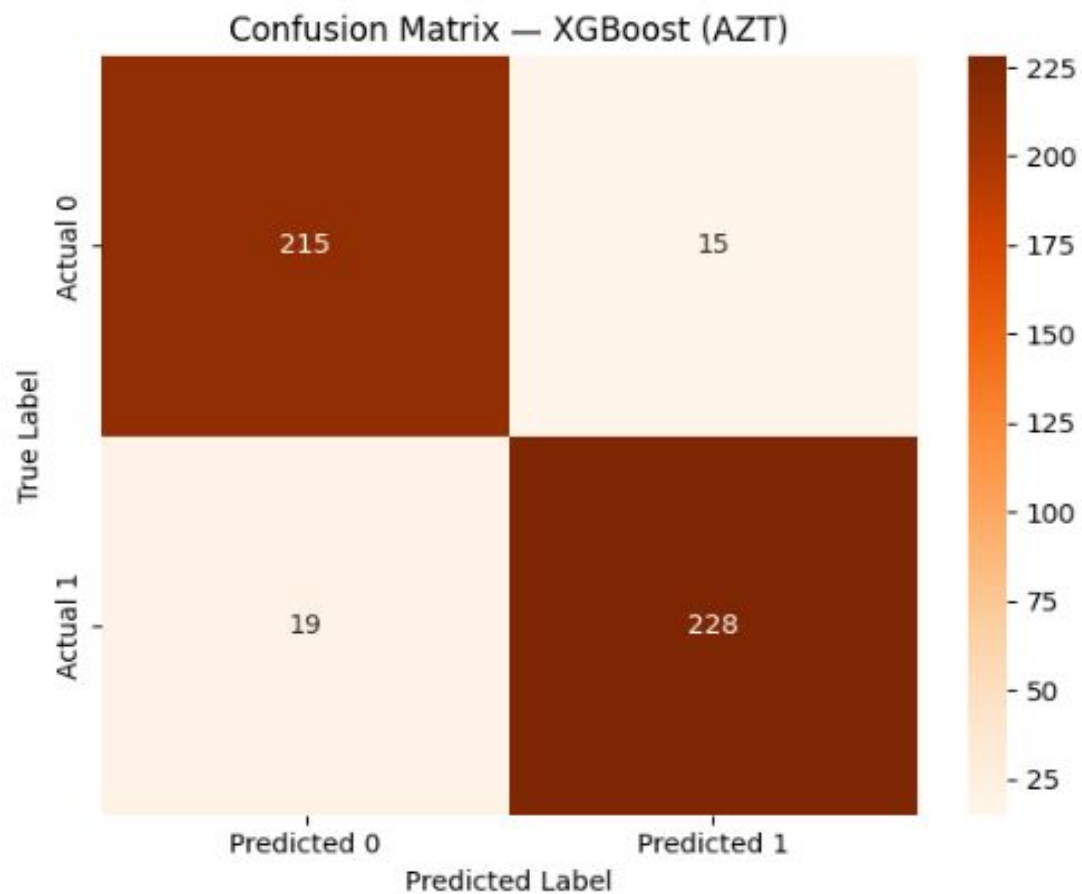
	precision	recall	f1-score	support
0	0.92	0.93	0.93	230
1	0.94	0.92	0.93	247
accuracy			0.93	477
macro avg	0.93	0.93	0.93	477
weighted avg	0.93	0.93	0.93	477

Evaluation metrics: **XGB-Accuracy**

--- XGBoost Test Classification Report ---

	precision	recall	f1-score	support
0	0.92	0.93	0.93	230
1	0.94	0.92	0.93	247
accuracy			0.93	477
macro avg	0.93	0.93	0.93	477
weighted avg	0.93	0.93	0.93	477

Confusion Matrix - XGBOOST



Precision- Recall & ROC AUC curves

Comparison (Macro -avg) - AZT

Model	Precision	Recall	F1 score
LR	0.92	0.92	0.92
RF	0.93	0.94	0.93
SVM	0.91	0.91	0.91
XGB	0.93	0.93	0.93

Model Ranking by MCC (AZT)

Rank	Model	MCC
1	RF	0.8704
2	XGB	0.8575
3	LR	0.8368
4	SVM	0.8195

Best model : RF